

GRADUATION PROJECT PRESENTATION - MODULE

Chrome Extension, GNOME Desktop Shell and Social Systems

Manifest V3 backgrounds, site blocking declarative rules, GJS status applets, and leaderboard point aggregates.

Presenter: Member 7

Core Focus: Extensions & Social Features

Menoufia University | Computer Science & Engineering Division

Chrome Extension MV3 Service Workers

- Cortex extends focus controls to the browser using a Chrome extension.
- The extension runs background service workers to monitor focus timer states.
- Service workers monitor alarms to toggle site-blocking configurations.
- [SCREENSHOT: Chrome settings listing developer extension status and active service worker]

```
// extension manifest MV3 JSON
{
  "manifest_version": 3,
  "name": "Cortex Extension",
  "permissions": ["declarativeNetRequest", "storage", "alarms"],
  "background": { "service_worker": "background.js" }
}
```

Dynamic site blocker declarativeNetRequest

- The extension blocks distracting sites during study sessions.
- We register dynamic rule sets using browser APIs.
- The script removes blocking rules automatically when the timer ends.
- [SCREENSHOT: Chrome browser displaying access blocked alert screens during focus]

```
// declarativeNetRequest rule registration
chrome.declarativeNetRequest.updateDynamicRules({
  addRules: [{
    id: 1, priority: 1, action: { type: "block" },
    condition: { urlFilter: "youtube.com", resourceTypes: ["main_frame"] }
  }]
});
```

GNOME Desktop Applet Status Indicator

- Cortex integrates focus indicators directly into the Linux status area.
- The widget button class extends GNOME panel menus.
- Dynamic labels display remaining focus session minutes.
- [SCREENSHOT: Linux desktop panel rendering active study applet widget]

```
// GNOME Applet label configuration
this.box = new St.BoxLayout();
this.label = new St.Label({ text: 'Cortex Focus: Idle' });
this.box.add_child(this.label);
```

GNOME Applet Focus Polling GJS

- The GNOME widget applet is written in Javascript for GNOME GJS.
- The applet checks local state files periodically to poll focus status.
- The applet updates panel label text and system icons to indicate active focus states.
- [SCREENSHOT: GNOME logs console debugging polling tick intervals]

```
// GJS File Reader polling loop
_pollFocusState() {
    let file = Gio.File.new_for_path('/tmp/cortex_focus.state');
    let [success, contents] = file.load_contents(null);
    // Parse coordinates and update UI elements...
}
```

Study Groups & Members Tables DDL

- The database layout stores group names, descriptions, and members.
- The schema links profiles using cascading foreign keys.
- We index membership tables to speed up group directories lookup.
- [SCREENSHOT: Group panel showing classmates name listings and workspace files]

```
-- Study groups schema DDL
CREATE TABLE study_groups (
  id UUID PRIMARY KEY,
  name TEXT NOT NULL,
  invite_code TEXT UNIQUE,
  created_by UUID REFERENCES profiles(id)
);
```

Workspace Invites & Dynamic validation

- The database validates alphanumeric invite codes before joining users.
- System prompts ask users to input group invite codes.
- We verify group codes in the database to register memberships.
- [SCREENSHOT: Popups screen asking for group invite pin input]

```
// Check group invite codes
const { data: group } = await supabase
  .from("study_groups")
  .select("id")
  .eq("invite_code", inputCode)
  .single();
```

Competitive Group Leaderboards

- Cortex calculates leaderboard positions using a point calculation query.
- Group members earn points by logging study hours and publishing notes.
- The leaderboard ranking query joins profile tables to display usernames.
- [SCREENSHOT: Students ranking leaderboards visual board dashboard]

```
-- Leaderboard point ranking query
SELECT
    user_id,
    SUM(points) as total_pts,
    ROW_NUMBER() OVER (ORDER BY SUM(points) DESC) as rank
FROM user_points
GROUP BY user_id;
```


Social Points Allocations Rules

- The points system rewards consistency and cooperative sharing.
- The points calculator prevents spam by applying daily frequency limits.
- Ranking logs update dynamically, keeping leaderboard positions updated.
- [SCREENSHOT: System configuration card detail points allocations rules]

User Action Type	Points Earned	Frequency Limit	System Tier Level
Log Pomodoro timer	10 points / hr	No daily limit	High consistency
Publish notes resource	5 points / note	Max 5 notes / day	Verified review
Complete habits log	2 points / log	Once daily per habit	Daily check
Accept friends request	1 point / link	Max 50 links	Social connections

Multi-Platform Extensions Sync Rates

- Students access study tools across web, browser, and desktop clients.
- API routes handle database queries, syncing metrics across devices.
- Security rate limiters manage traffic to ensure stable performance.
- [SCREENSHOT: Extension configurations panel showing status indicator toggles]

